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Marketing

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Report Contents

Executive Summary

Tata Motors Group

The Challenge

The Role of CLV

Segmentation, Targeting, Positioning

Product Targeting

Price Optimisation

Concluding Remarks

Bibliography

Executive Summary

This report demonstrates how data analytics can augment traditional marketing frameworks, to better solve emerging marketing challenges. The case study is the Tata Motors Group, which, like many other automotive manufacturers, is facing slumping demand due to both to the Covid-19 pandemic and more general shifts in customer tastes.

After identifying this challenge, the report explains the usefulness of customer lifetime value analysis, and CLV estimations for consumers of durable goods could be improved using multiple-regression analysis. Then, the report outlines the segmentation-targeting-positioning strategy, exploring how clustering and regression analysis could refine how the firm segments the market. Finally, I explain how marketing strategies relating to product and price can be guided by data to better target specific customer segments. Key ideas here include using product-performance data and unstructured machine learning to improve the firm's understanding of current and emerging customer needs.

Tata Motors Group

This essay will focus on a Covid-related challenge facing Tata Motors Group (TMG), with my analysis rooted in TMG's context as part of the wider Tata Group conglomerate. Interestingly, the Tata Group blends the "branded-house" and the "house of brands" approaches. For example, Tata Motors (TM), Chemicals and Steel are part of a branded 'Tata' house, which Jaguar Land Rover (JLR) and Indian Hotels Company exist outside of. This gives TMG scope for a flexible marketing approach, as it can target multiple segments using its TM and JLR brands, without needing to maintain a consistent message between the two. This premise underscores much of my analysis.

The Challenge

As detailed in TMG's 2020 Annual Report, Covid-19 has contributed to declining automotive sales by stagnating incomes. Further, lockdown mandates have encouraged customers to postpone purchases of automotives, due to lower sentiment and the reduced accessibility of products. These factors have contributed to TMG's -0.2% EBIT margin in 2020, a loss of £422m.

Further, the environmental awareness spurred by lockdowns throughout the pandemic is accelerating trends toward shared and electric vehicles (Carrington, 2021). These preference shifts affect both primary and secondary demand. Primary demand for conventional automotives has fallen due to declining incomes and increasing environmental awareness, supported by tightening emissions regulations. The magnitude of the shifts is not constant across brands, due to differentials in income and price elasticity of demand, and differential abilities to meet shifting tastes. For example, firms like Tesla, who possess some first-mover advantages for popular electric vehicles, may be better positioned to benefit from this trend in customer preferences.

This essay will explore how data-driven marketing can help solve TMG's customer problem by creating better alignment between product capabilities and evolving customer needs. My solutions use data on customer lifetime valuations (CLV) to inform a segmentation-targeting-positioning (STP) strategy which can increase profit margins. This is achieved by modifying products to increase value, which is then communicated through promotions and captured by price optimisation, all informed by data.

The Role of CLV

Customer lifetime valuation (CLV) is the net present value of the current and future cash flows attributed to a relationship with a customer. CLV is obtained through the calculation:

$$CLV = \sum \frac{(M_t - C_t)}{(1+i)^t} r^t$$

Calculating CLV is useful to TMG for several reasons. First, if classifies its customers by CLV, TMG will derive an upper-bound of acquisition costs for each segment it serves (Farris, 2010). Intuitively, Western consumers of luxury JLR vehicles will likely have a higher CLV than Indian consumers of cost-leading TM vehicles. This may be because JLR margins ($M_t - C_t$) or retention rates (r_t) are higher. Understanding the exact differentials in CLV between segments will inform the allocation of TMG's marketing resources, such that for every customer $CLV >$ acquisition costs. This is the key takeaway of CLV; marketing campaigns must add value, not destroy it. The Wayfair case highlights the importance of this principle; after McCarthy and Fader's (2017) customer-based corporate valuation gave a \$10.24 stock price, investors realised Wayfair's stock was grossly overvalued on the market and so the stock tumbled.

Secondly, as mentioned above, aggregating the CLVs of specific product lines and families allows us to understand the current value of the brands producing them. This customer-based corporate valuation (CBCV) is a bottom-up approach which recognises that not all customers create equal revenue for a firm. This is particularly important for TMG due to CLV differentials between its brands. After using CBCV to gauge its most and least valuable divisions, TMG could potentially rationalise business units and products with poor performance outlooks. This roots cost-saving measures in data that provides deeper insights than revenue figures, thus reducing the role of bias in decision-making.

As such, TMG should collect data on margins, retention rates, acquisition costs and discount rates (i). The firm will likely already have data on the profit margins for each of its products. The discount rate should also be accessible, as this is simply the firm's weighted average cost of capital (WACC). Calculating acquisition costs could be more difficult for TMG; there is large potential for cross-selling within the branded house of Tata. In these situations, how should TMG treat acquisition costs? The onus should be on Tata Group to set out a consistent methodology for apportioning acquisition costs between its verticals. Retention rates may also be hard to observe because TMG's non-contractual offering means customers do not need to communicate consumption discontinuation (McCarthy and Fader, 2017). The recommended approach is to compare the recency of last purchase with the average purchase frequency for the product category, which will be low for durable goods.

However, here TMG could seek to understand each customer's purchase cycle using data-driven approaches, for example by creating multiple-regression models that use variables like customer income, recency of last purchase, family size, macroeconomic conditions etc. to estimate when the next purchase will be. This could increase the accuracy of retention rate estimations, and can align the firm's allocation of marketing resources with purchasing cycles, i.e. by targeting customers more heavily when the regression model predicts a purchase will soon be made.

It is important to note that as well as informing marketing strategies, CLV is also a function of them. This brings us to a third use of CLV for TMG; if the firm can understand the underlying drivers of margins and retention rates, then increase these via its marketing mix, it can raise its average CLV. How might this be achieved?

Segmentation, Targeting, Positioning

Segmentation is the process of dividing the customer base into distinct groups of homogenous or clustered preferences (Kotler, 2016). Under the STP framework, firms should then target segments based on attractiveness (Hanlon, 2021) before positioning its products to meet the needs of the selected customers. As already mentioned, TMG competes in many segments, offering passenger vehicles in premium (Jaguar) and cost-leading (Tata Nano) segments, as well as commercial vehicles. My strategy would begin by using data-driven marketing to refine TMG's segmentation practices, perhaps redefining segment boundaries or revaluing their attractiveness to TMG.

Kotler (2019) posits that needs-based segmentation involves identifying segments, assessing their attractiveness, then implementing a targeted marketing mix. As explained, CLV analysis informs the first two steps, highlighting the value of this data to TMG. Further, Wu and Lin (2005) show that unsupervised clustering analysis can aid segmentation by grouping customers based on multiple observed variables. Alternatively, TMG could use supervised regression analyses to predict purchase likelihoods of various groups based on a range of data points in demographic and psychographic categories. As well as helping TMG define and value segments, these processes could also help the firm achieve cost-savings by directing promotional activities towards marginal customers (Blake, 2015).

After defining segments and considering their attractiveness, TMG must consider its resource constraints and business objectives before settling on target segments (Kotler, 2019). This is important because even if a segment is attractive, TMG may lack the resources and capabilities to meet its needs, or competitors may be better positioned to do so. After selecting segments, TMG should implement a specific marketing mix, honing its "4 Ps" strategy for current and prospective target segments. Guided by data insights, a targeted marketing mix will allow TMG to deliver more value to customers, which, if captured by its pricing strategy, will allow it to increase its customer-based corporate valuation.

Product Targeting

After identifying target segments, TMG should leverage new technologies to collect and analyse data so it can better understand shifting customer needs. Using these insights to guide product modification and development will allow TMG to deliver more value in this dynamic market.

To illustrate, sensory technology is allowing ever-increasing amounts of product-performance data to be collected. Such technologies are becoming more cost-effective, thus increasingly used in sectors like aerospace to determine how products are used (Duncan, 2014). Tata Steel has previously used these data analytics *internally* to train a machine learning algorithm which honed its production process, decreasing defect rates (Kumar, 2021). TMG could combine this approach with *external* analytics, using sensory technology to learn about the frequency of use of different features in its autos, and the conditions the auto is operating in. Going further, conducting regression analysis to identify correlations between external environments and product features used would allow TMG to understand the *specific contexts* leading to unfulfilled customer needs, unneeded product capabilities and key value drivers. These data insights can be leveraged to cut costs and increase value, by shifting resources from unrequired to required capabilities.

TMG could also focus on data external to the consumption interaction. For example, Natural Language Processing techniques can analyse unstructured text data such as online reviews, news, and social media posts, deriving customer valence and topics discussed (Deng, 2021). In contrast to product-performance data, this could give TMG information about potential *future* customer needs, which could create a competitive edge in an increasingly dynamic automotive market.

Building these insights into product development to better align to customer needs should increase customers' willingness to pay. This gives TMG scope to set higher prices; in a value-based pricing strategy, price is a mark-down on customers' willingness to pay (Hinterhuber, 2004). However, is it possible to again use data-driven marketing to optimise TMG's pricing strategy?

Price Optimisation

Data-driven marketing can increase product quality whilst making cost-savings. Achieving better alignment to shifting customer needs will give TMG scope to raise prices, despite faltering economic growth. Indeed, Farris and Reibstein (1979) found that brands with high quality and advertising spend can charge higher prices. This alludes to an important qualifying point: to make price increases palatable to customers, TMG must ensure this increased value is communicated through promotional activities.

According to Kotler (2016), pricing decisions should select an objective, determine demand, estimate costs, analyse competitors, then select the pricing method. These steps can be optimised with data-driven marketing. For example, Dube (2017) shows how machine learning can derive individual demand estimations from data inputs, then use scalable price targeting to maximise revenues. Further, Weed (2020) explains how hyper-dynamic pricing is made possible by machine learning rapidly analysing data including news stories, weather predictions and Google searches to infer demand and analyse competitors.

Whilst first-degree price discrimination (Dube) may not be possible for TMG, customer-centred prices would follow Oliver Wyman's (2017) recommendation to automakers. It believes the industry needs to shift to differentiated and dynamic price listings, as opposed to crude cost-plus pricing strategies – I would be inclined to agree. Additionally, the firm found that 30% of a car's list price reflects distribution costs, including dealer margins. I believe here there is further scope to leverage data analytics, to redesign distribution channel structures thus avoid unnecessary costs.

Concluding Remarks

Overall, Tata Motors Group is challenged by falling sales due to shifting customer tastes, accelerated by Covid-19. Aside from lowering incomes and causing lockdowns, it has strengthened trends towards environmental awareness and shared vehicles with integrated technology.

To tackle this problem, I would apply the segmentation-targeting-positioning framework. The first two steps would involve leveraging data collected on CLVs to inform the allocation of marketing resources, and leveraging data on customer needs and product-performance to increase the value of TMG's offerings. Automotive manufacturers like TMG likely already conduct similar analyses, but the constant evolution of data and computing means new techniques are always emerging. Tools such as regression analysis and unstructured machine learning can identify patterns in collected data, which are then fed back to manufacturing to improve the alignment between product capabilities and target customer needs. This proposed feedback loop underscores the role of data-driven marketing in guiding other business functions. After improving this alignment, I would use targeted pricing strategies to capture the additional product value, discriminating based on collected customer data. TMG would also need to explore the role of targeted promotions in ensuring this additional value is converted into a higher willingness to pay amongst its customers.

My analysis is strengthened considering that TMG is part of the wider Tata Group. The Tata branded house creates strong potential for cross-selling between verticals, driving down customer acquisition costs, potentially reducing TMG's barriers to entry into new segments. Simultaneously, the house of brands approach seen when comparing the JLR and TM brands gives TMG flexibility to pursue a multi-dimensional STP strategy. Targeting multiple attractive segments will help TMG further diversify its risk exposure, both geographically and through different demographic groups.

Bibliography

- Blake, T., Nosko, C. and Tadelis, S. (2015). 'Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment', *Econometrica*, 83 (10), pp. 155-174.
- Carrington, D. (2020). 'Global sales of electric cars accelerate fast in 2020 despite pandemic', *The Guardian*, 19 January [online]. Available at: <https://www.theguardian.com/environment/2021/jan/19/global-sales-of-electric-cars-accelerate-fast-in-2020-despite-covid-pandemic#:~:text=Global%20sales%20of%20electric%20cars%20accelerated%20fast%20in%202020%2C%20rising,almost%20500%2C000%2C%20followed%20by%20Volkswagen>. (Accessed 24/04/20).
- Deng, Y. (2021).
- Dube, J-P. and Misra, S. (2017). 'Scalable Price Targeting', *National Bureau of Economic Research*, Working Paper No. w23775.
- Duncan, E. and Ritter, R. (2014). 'Next frontiers for Lean', *McKinsey Quarterly*, 1 February [online]. Available at: <https://www.mckinsey.com/business-functions/operations/our-insights/next-frontiers-for-lean> (Accessed 19/04/21).
- Farris, P. and Reibstein, D. (1979). 'How Prices, Ad Expenditures, and Profits Are Linked', *Harvard Business Review*, November 1979 issue.
- Farris, P., Bendle, N., Pfeifer, P., Reibstein, D. (2010). *Marketing Metrics: The Definitive Guide to Measuring Market Performance*, Upper Saddle River, New Jersey.
- Hanlon, A. (2021). 'The segmentation, targeting, positioning (STP) marketing model', *Smart Insights*, 26 January [online]. Available at: <https://www.smartinsights.com/digital-marketing-strategy/customer-segmentation-targeting/segmentation-targeting-and-positioning/> (Accessed 20/04/21).
- Hinterhuber, A. (2004). 'Towards value-based pricing – An integrative framework for decision making', *Industrial Marketing Management*, 33 (8), pp. 765-778.
- Joas, A., Brandt, F. and Daffner, R. (2017). 'Achieving automotive pricing excellence', *Oliver Wyman*. Available at: <https://www.oliverwyman.com/content/dam/oliver-wyman/v2/publications/2017/jun/Art9-Achieving-automotive.pdf>
- Kotler, P. and Keller, K. 2016. *Marketing management 3rd ed.*, Pearson.
- Kotler, P. and Keller, K. 2019. *Marketing management: European Edition*, Pearson.
- Kumar, R., Dubashi, N. and Gupta, R. (2021). 'How Tata Steel Tapped The Value of Data', *Tata.com*, April. Available at: <https://www.tata.com/newsroom/business/tata-steel-kalinganagar-data-analytics-mckinsey> (Accessed 19/04/21).
- McCarthy, D. and Fader, P. (2017). 'Subscription Businesses Are Booming. Here's How to Value Them', *Harvard Business Review*, December 2017 issue.
- McCarthy, D. and Fader, P. (2018). 'Customer-Based Corporate Valuation for Publicly Traded Noncontractual Firms', *American Marketing Association*, 55 (5), pp. 617-635.
- Tata Motors Group Ltd. (2020). 75th *Integrated Annual Report*. Available at: <https://investors.tatamotors.com/financials/75-ar-html/index.html> (Accessed 20/04/21).
- Weed, J. (2020). 'In the Race for Cheap Airfare, It's You vs. the Machine', *New York Times*, 29 January. Available at: <https://global-factiva-com.proxy.jbs.cam.ac.uk/ga/default.aspx> (Accessed 19/04/21).
- Wu, J. and Lin, Z. (2005). 'Research on Customer Segregation Model by Clustering', *ICEC '05: Proceedings of the 7th international conference on Electronic commerce*, pp. 316-318.